COMP 3721 **Assignment 4**

Note:

- Submit a single PDF file.
- Answer the questions according to textbook, lecture notes and class discussions, not ChatGPT or Google.

The beginning bytes of two IP packets are shown as follows. Data are in hexadecimal.

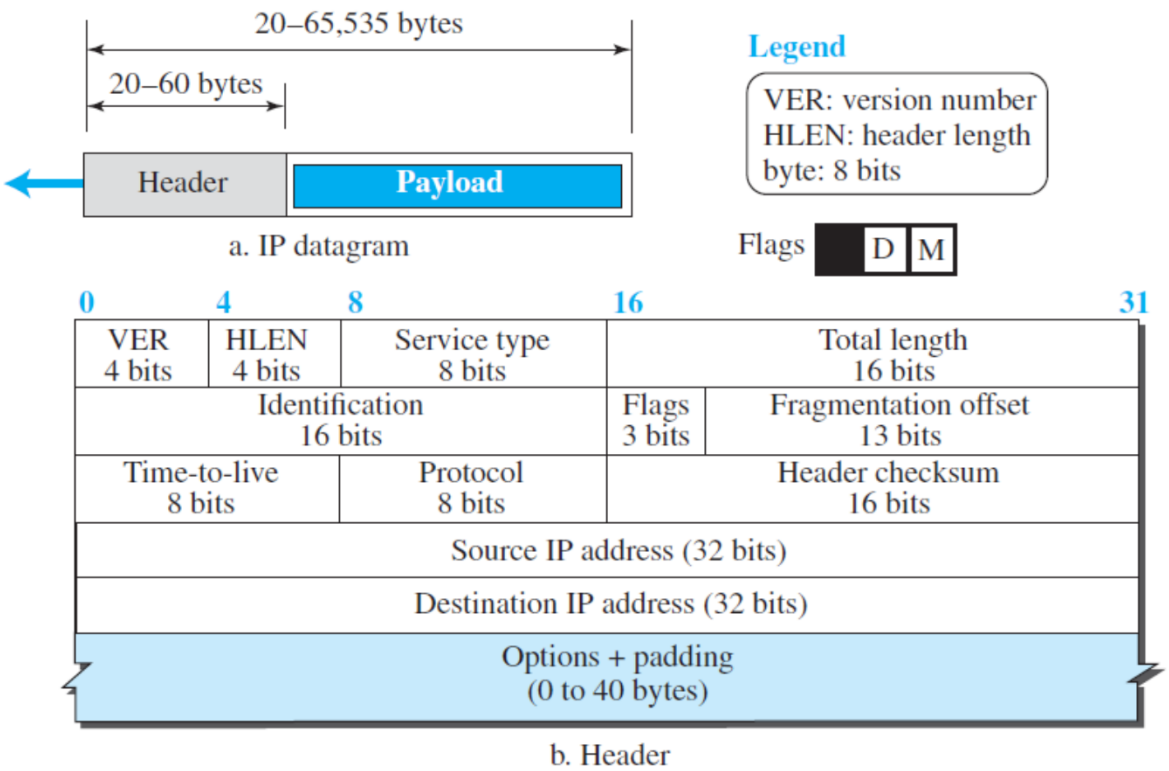
Packet 1:

```
45 00 05 DC
A5 E5 20 00
80 01 00 00
51 1C 9A 62
17 D5 E6 9A …
```

Packet 2:

```
45 00 05 DC
A5 E5 00 B9
80 01 00 00
51 1C 9A 62
17 D5 E6 9A …
```

The IP datagram format is shown below for your reference.



Flags (3-bit field):

R: Reserved. Always 0.

D: If set to 1, the datagram must not be fragmented

M: 0 for the last or only fragment

Some protocol values:

ICMP: 01

TCP: 06 UDP: 17

Q1:

Find the following for packet 1.

(a) Version number

4

(b) Header length (in bytes)

5 * 4 = 20 bytes

(c) Confirm that service type is 0.

00 (hex) = 0000 0000 (binary) = 0(decimal)

(d) Total length of IP packet (in bytes)

05 DC=
(0*16^3) + (5*16^2) + (13*16^1) + (12 * 16^0) =
0 + 1280 + 208 + 12 = 1500 bytes

(e) Identification number (in decimal)

A5 E5 =
(10*16^3) + (5*16^2) +(14*16^1) + (5*16^0)=
(40960 + 1280 + 224 + 5)=
42469

(f) Can we fragment this IP packet?

Yes
20 (hex) = 0010 0000 (binary)
First 3 bits = 0 0 1
0 = R = 0
0= D = 0 = fragment allowed
0= M = 1

(g) Are there more fragments following this IP packet?

Yes(M = 1)
20 (hex) = 0010 0000 (binary)
First 3 bits = 0 0 1
0 = R = 0
0= D = 0 = fragment allowed
0= M = 1

(h) Fragmentation offset of this packet (in decimal)

20 (hex) = 0010 0000
00 (hex) = 0000 0000
0010 0000 0000 0000
Remove first 3 bits(flags) = 001
Fragment offset(13 bits) = 0 0000 0000 0000 = 0
Fragment offset = 0

(i) Time-to-live (in decimal)

80 (hex) =
(8*16^1) + (0*16^0) =
(128 + 0) =
TTL = 128

(j) Protocol (e.g., ICMP, TCP, UDP, etc.)

80 01
01 = ICMP

(k) Check that the header checksum is 0x0000 (not calculated). Calculate the checksum according to the lecture notes. Write the answer in hexadecimal.

```
4 5 0 0
0 5 D C
A 5 E 5
2 0 0 0
8 0 0 1
0 0 0 0
5 1 1 C
9 A 6 2
1 7 D 5
E 6 9 A

0+12+5+0+1+0+12+2+5+10 = 47= 0x2F
2+0+13+14+0+0+1+6+13+9 =58= 0x3A
3+5+5+5+0+0+0+1+10+7+6= 42= 2A
2+4+0+10+2+8+0+5+9+1+14= 37

4 5 0 0
0 5 D C
A 5 E 5
2 0 0 0
8 0 0 1
0 0 0 0
5 1 1 C
9 A 6 2
1 7 D 5
E 6 9 A
———————
Sum= 37AAF
Wrapped sum = 7AB2
Checksum = FFFF-7AB2= 0x854D
```

(l) Source IP address (in dotted decimal)

```
51 1C 9A 62
51 (hex) = (5*16^1) + (1*16^0)=81(decimal)
1C (hex) = (1*16^1) + (12*16^0)= 28
9A (hex) = (9*16^1) + (10*16^0)= 154
62 (hex) = (6*16^1) + (2*16^0)= 98
```

Source IP address (dotted decimal)=81.28.154.98

(m) Destination IP address (in dotted decimal)

```
17 D5 E6 9A
17 (hex) = (1*16^1) + (7*16^0)= 23
D5 (hex) = (13*16^1) + (5*16^0)= 213
E6 (hex) = (14*16^1) + (6*16^0)= 230
9A (hex) = (9*16^1) + (A*16^0)= 154
```

IP address (in dotted decimal)= 23.213.230.154

Q2:

Find the following for packet 2.

(a) Version number

4

(b) Header length (in bytes)

5 * 4 = 20 bytes

(c) Confirm that service type is 0.

00 (hex) = 0000 0000 (binary) = 0(decimal)

(d) Total length of IP packet (in bytes)

05 DC=
(0*16^3) + (5*16^2) + (13*16^1) + (12 * 16^0) =
0 + 1280 + 208 + 12 = 1500 bytes

(e) Identification number (in decimal)

A5 E5 =
(10*16^3) + (5*16^2) +(14*16^1) + (5*16^0)=
(40960 + 1280 + 224 + 5)=
42469

(f) Can we fragment this IP packet?

Yes
00 (hex) = 0000 0000 (binary)
First 3 bits = 0 0 0
0 = R = 0
0= D = 0 = fragment allowed
0= M = 0

(g) Are there more fragments following this IP packet?

No(M = 0)
00 (hex) = 0000 0000 (binary)
First 3 bits = 0 0 0
0 = R = 0
0= D = 0 = fragment allowed
0= M = 0

(h) Fragmentation offset of this packet (in decimal)

00 (hex) = 0000 0000
B9 (hex) = 1011 1001
0000 0000 1011 1001
Remove first 3 bits(flags) = 000
Fragment offset(13 bits) = 0000010111001 = 185
Fragment offset = 185

(i) Time-to-live (in decimal)

80 (hex) =
(8*16^1) + (0*16^0) =
(128 + 0) =
TTL = 128

(j) Protocol (e.g., ICMP, TCP, UDP, etc.)

80 01
01 = ICMP

(k) Check that the header checksum is 0x0000 (not calculated). Calculate the checksum according to the lecture notes. Write the answer in hexadecimal.

```
4 5 0 0
0 5 D C
A 5 E 5
0 0 B 9
8 0 0 1
0 0 0 0
5 1 1 C
9 A 6 2
1 7 D 5
E 6 9 A

0+12+5+9+1+0+12+2+5+10 = 56= 0x38
3+0+13+14+11+0+0+1+6+13+9 =70= 0x46
4+5+5+5+0+0+0+1+10+7+6= 43= 0x2B
2+4+0+10+0+8+0+5+9+1+14= 53= 0x35
4 5 0 0
0 5 D C
A 5 E 5
0 0 B 9
8 0 0 1
0 0 0 0
5 1 1 C
9 A 6 2
1 7 D 5
E 6 9 A
————————
Sum= 35B68
Wrapped sum = 5B6B
Checksum = FFFF-5B6B= 0xA494
```

(l) Source IP address (in dotted decimal)

```
51 1C 9A 62
51 (hex) = (5*16^1) + (1*16^0)=81(decimal)
1C (hex) = (1*16^1) + (12*16^0)= 28
9A (hex) = (9*16^1) + (10*16^0)= 154
62 (hex) = (6*16^1) + (2*16^0)= 98
```

Source IP address (dotted decimal)=81.28.154.98

(m) Destination IP address (in dotted decimal)

```
17 D5 E6 9A
17 (hex) = (1*16^1) + (7*16^0)= 23
D5 (hex) = (13*16^1) + (5*16^0)= 213
E6 (hex) = (14*16^1) + (6*16^0)= 230
9A (hex) = (9*16^1) + (A*16^0)= 154
```

IP address (in dotted decimal)= 23.213.230.154

Q3:

(a) Assuming that these two packets are the fragments that belong to the same packet (note the identification numbers), which packet is the first fragment and which is the last? Why?

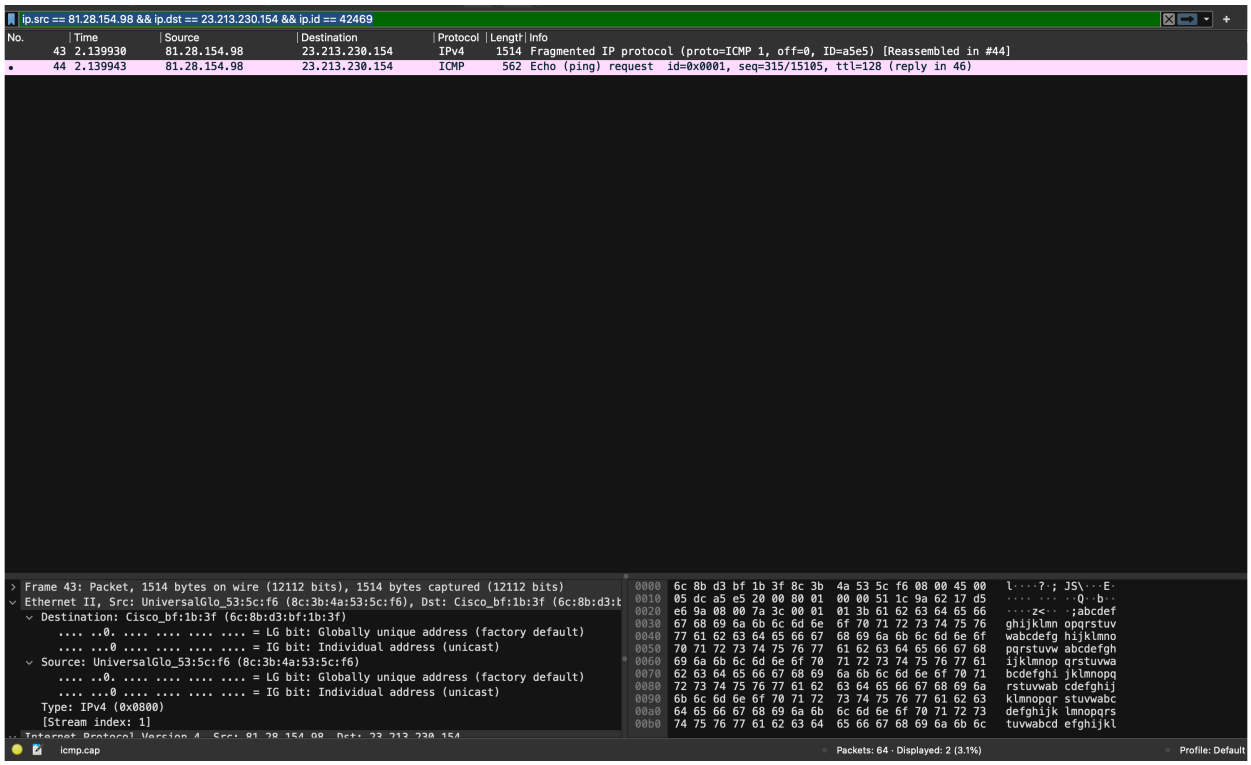
Packet 1 is the first fragment because Packet 1 has fragment offset = 0 and packet 1 has M = 1, so more fragments follow

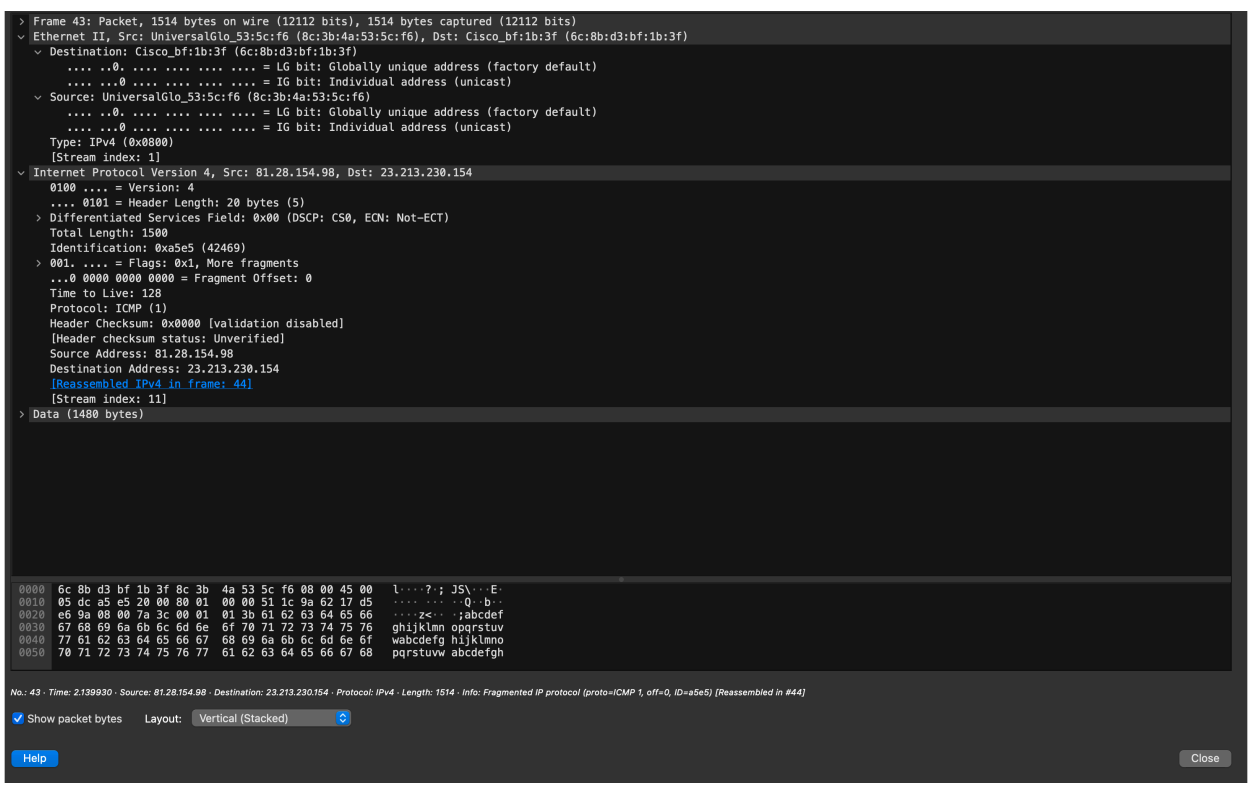
Packet 2 is the last fragment because Packet 2 has fragment offset = 185 and packet 2 has M = 0, so this is the last fragment

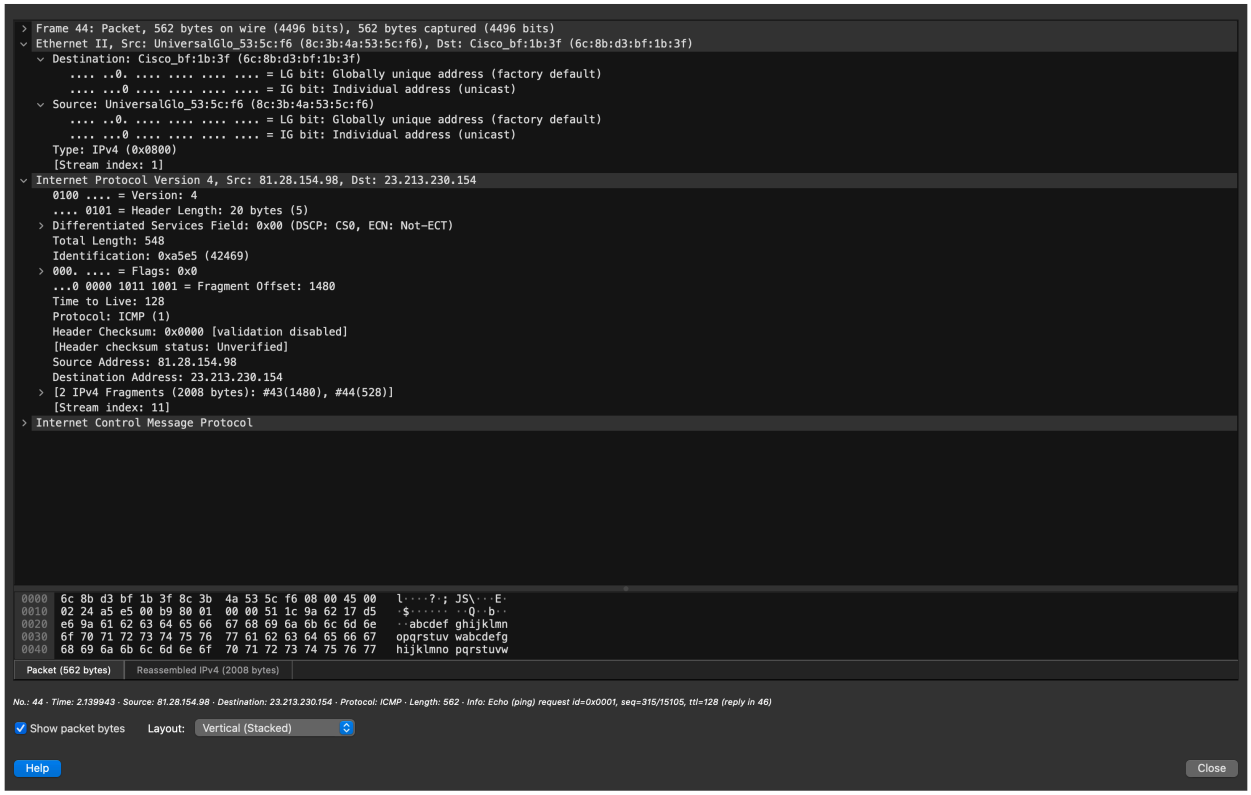
(b) Using Wireshark, find the above packets in the given "icmp.cap" file. Show the filter used in Wireshark, and identify their frame numbers (e.g., packet 1 is frame 64 and packet 2 is frame 100).
(Note 1: Use the answer(s) you have in Q1 and Q2 to construct suitable filter(s).)
(Note 2: Frame number is the first column in Wireshark, denoted as "No.".)

ip.src == 81.28.154.98 && ip.dst == 23.213.230.154 && ip.id == 42469

Packet 1 = frame 43
Packet 2 = frame 44


ip.src == 81.28.154.98 && ip.dst == 23.213.230.154 && ip.id == 42469
No. Time Source Destination Protocol Length Info
43 2.139930 81.28.154.98 23.213.230.154 IPv4 1514 Fragmented IP protocol (proto=ICMP 1, off=0, ID=a5e5) [Reassembled in #44]
44 2.139943 81.28.154.98 23.213.230.154 ICMP 562 Echo (ping) request id=0x0001, seq=315/15105, ttl=128 (reply in 46)
Frame 43: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)
Ethernet II, Src: UniversalGlo_53:5c:f6 (8c:3b:4a:53:5c:f6), Dst: Cisco_bf:1b:3f (6c:8b:d3:b
Destination: Cisco_bf:1b:3f (6c:8b:d3:bf:1b:3f)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: UniversalGlo_53:5c:f6 (8c:3b:4a:53:5c:f6)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 1]
0000 6c 8b d3 bf 1b 3f 8c 3b 4a 53 5c f6 08 00 45 00
0010 05 dc a5 e5 20 00 80 01 00 00 51 1c 9a 62 17 d5
0020 e6 9a 08 00 7a 3c 00 01 01 3b 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f
0050 70 71 72 73 74 75 76 77 61 62 63 64 65 66 67 68
0060 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 77 61
0070 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71
0080 72 73 74 75 76 77 61 62 63 64 65 66 67 68 69 6a
0090 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 77 61 62 63
00a0 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73
00b0 74 75 76 77 61 62 63 64 65 66 67 68 69 6a 6b 6c
icmp.cap
Packets: 64 · Displayed: 2 (3.1%)
Profile: Default


Frame 43: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)
Ethernet II, Src: UniversalGlo_53:5c:f6 (8c:3b:4a:53:5c:f6), Dst: Cisco_bf:1b:3f (6c:8b:d3:bf:1b:3f)
Destination: Cisco_bf:1b:3f (6c:8b:d3:bf:1b:3f)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: UniversalGlo_53:5c:f6 (8c:3b:4a:53:5c:f6)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 1]
Internet Protocol Version 4, Src: 81.28.154.98, Dst: 23.213.230.154
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0xa5e5 (42469)
001. .... = Flags: 0x1, More fragments
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 81.28.154.98
Destination Address: 23.213.230.154
[Reassembled IPv4 in frame: 44]
[Stream index: 11]
Data (1480 bytes)
0000 6c 8b d3 bf 1b 3f 8c 3b 4a 53 5c f6 08 00 45 00
0010 05 dc a5 e5 20 00 80 01 00 00 51 1c 9a 62 17 d5
0020 e6 9a 08 00 7a 3c 00 01 01 3b 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f
0050 70 71 72 73 74 75 76 77 61 62 63 64 65 66 67 68
No.: 43 · Time: 2.139930 · Source: 81.28.154.98 · Destination: 23.213.230.154 · Protocol: IPv4 · Length: 1514 · Info: Fragmented IP protocol (proto=ICMP 1, off=0, ID=a5e5) [Reassembled in #44]
Show packet bytes
Layout:
Vertical (Stacked)
Help
Close



— End of Assignment 4 —